Section 4: Experiment and Discussion

4.1 Experimental Setup

All experiments are conducted using RAS-SDNCloudSim executed on a standard workstation.

Three datasets of increasing complexity are evaluated (Mini, Small, Medium).

Table 1: Dataset configurations

Dataset	Hosts	VMs	Workloads	Congestion design
Mini	4	4	50	Bottleneck: 50 Mbps vs. 800 Mbps
Small	6	8	100	Asymmetric Core: 80 Mbps vs. 300 Mbps
Medium	12	20	500	Fat-Tree Core: 200–500 Mbps multi-path

Table 2: Physical host power model (EnhancedHostEnergyModel)

Extends energy estimation to three dimensions (CPU, RAM, bandwidth).

P(t) = P_idle + alpha_cpu*u_cpu + alpha_ram*u_ram + alpha_bw*u_bw

(Parameters: P_idle=100W, alpha_cpu=1.0, alpha_ram=0.2, alpha_bw=0.1)

4.2 Results and Analysis

4.2.1 End-to-End Packet Delay

BLA consistently reduces delay under LFF and LWFF. Under MFF, BLA produces zero gain, confirming the measured improvements are exclusively due to routing (falsifiability criterion).

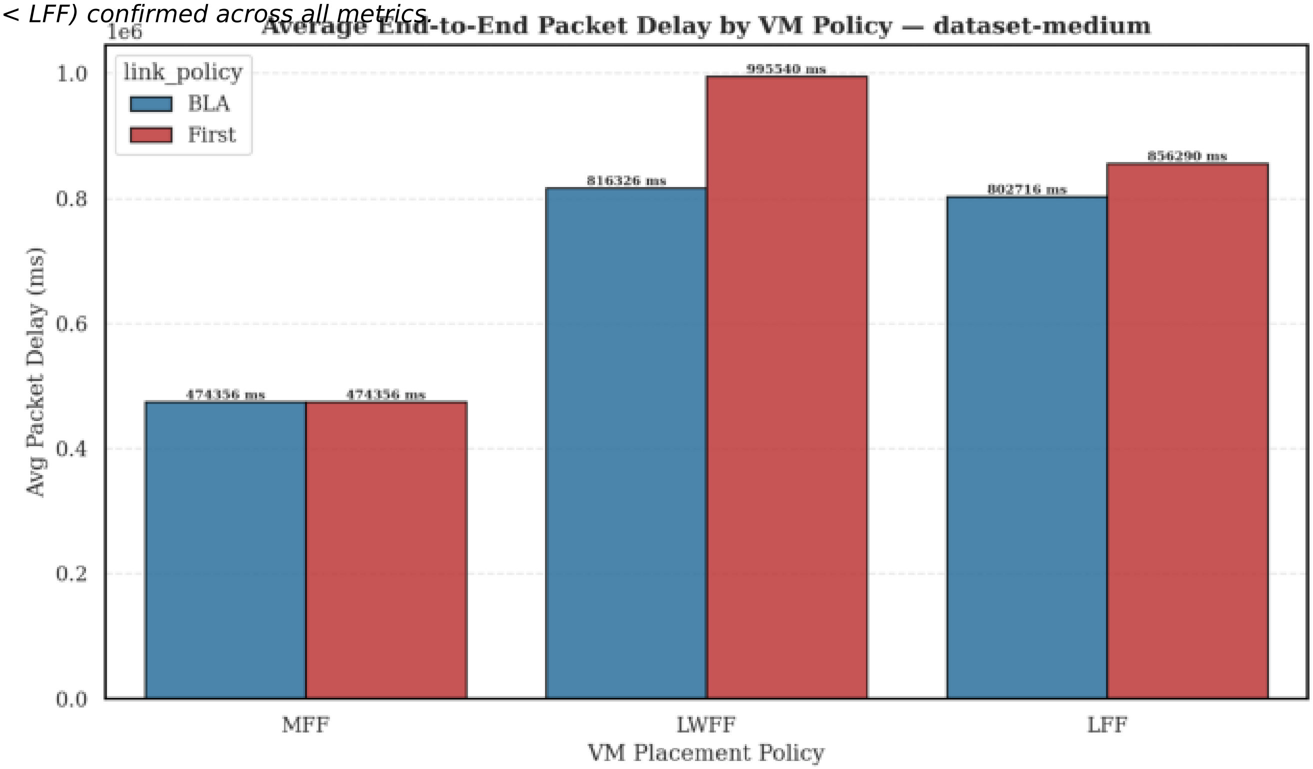
Table 4: Average packet delay (s) and BLA gain (%)

Dataset	VM Policy	SFL (s)	BLA (s)	Gain delay	Gain queue
Mini	LFF	48.4	37.8	-21.9%	-20.8%
Medium	LWFF	995.5	816.3	-18.0%	-19.1%

Remarks: The median gap is most diagnostic: at Medium scale, BLA median is 37% lower than First-Fit, indicating BLA prevents severe tail-latency degradation under sustained load.

FF << LWFF < LFF) confirmed across all metrics.

Figure 1: Average Packet Delay (Medium)



4.2.3 Queuing Analysis

4.2.3 Queuing Delay: Empirical Validation of the M/M/1 Model

Queuing delay is the most direct metric for validating the BLA mechanism.

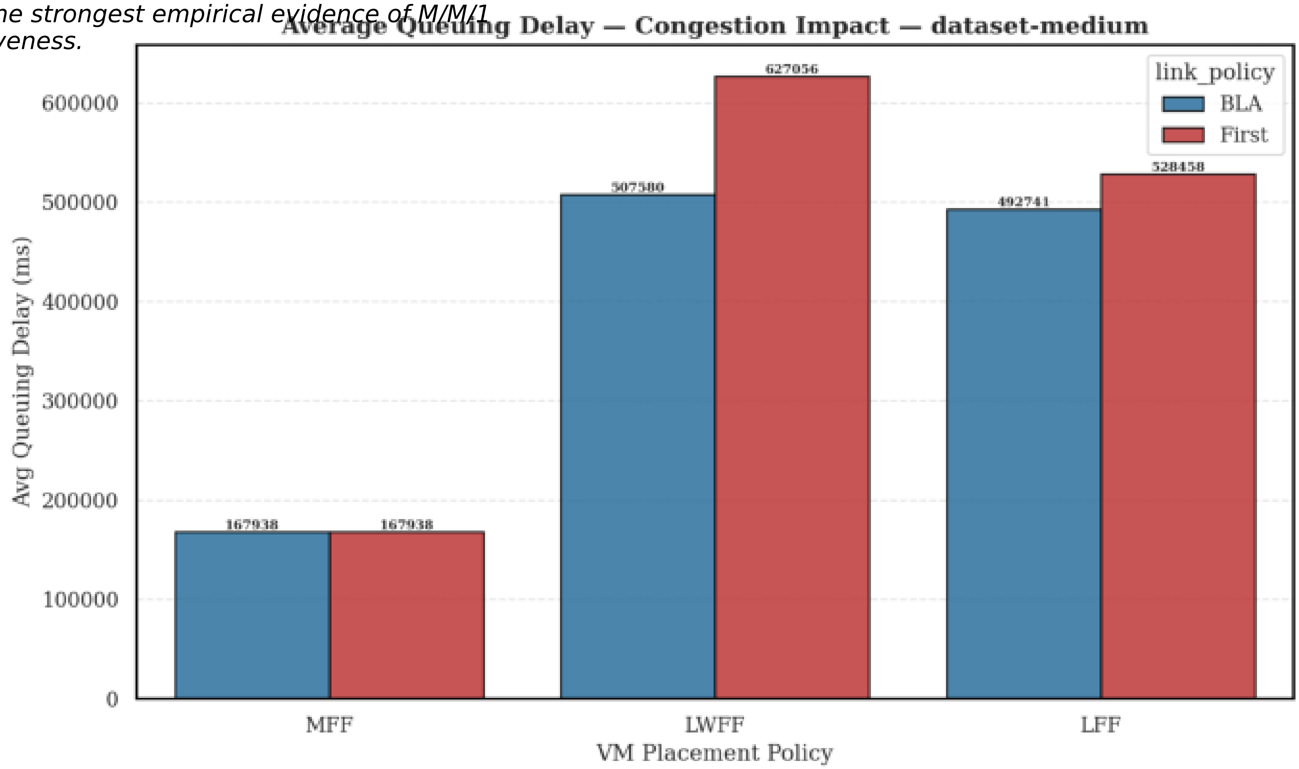
BLA does not simply reroute packets; it actively prevents congestion buildup at the queuing layer.

At Medium scale, BLA cuts queuing delay by 11.7% to 19.1%, demonstrating that bandwidth-aware

path selection avoids bottleneck formation before queues saturate.

Figure 2: Average Queuing Delay (Medium)

g by 19.1%, the strongest empirical evidence of M/M/1 effectiveness.



4.2.4 Energy Analysis

4.2.4 Energy Consumption
BLA accelerates workload completion by resolving queuing bottlenecks earlier. Temporal Compression Effect: E = P * Delta_t. Reduction in queuing time leads to shorter host active states and earlier entry to idle power modes.

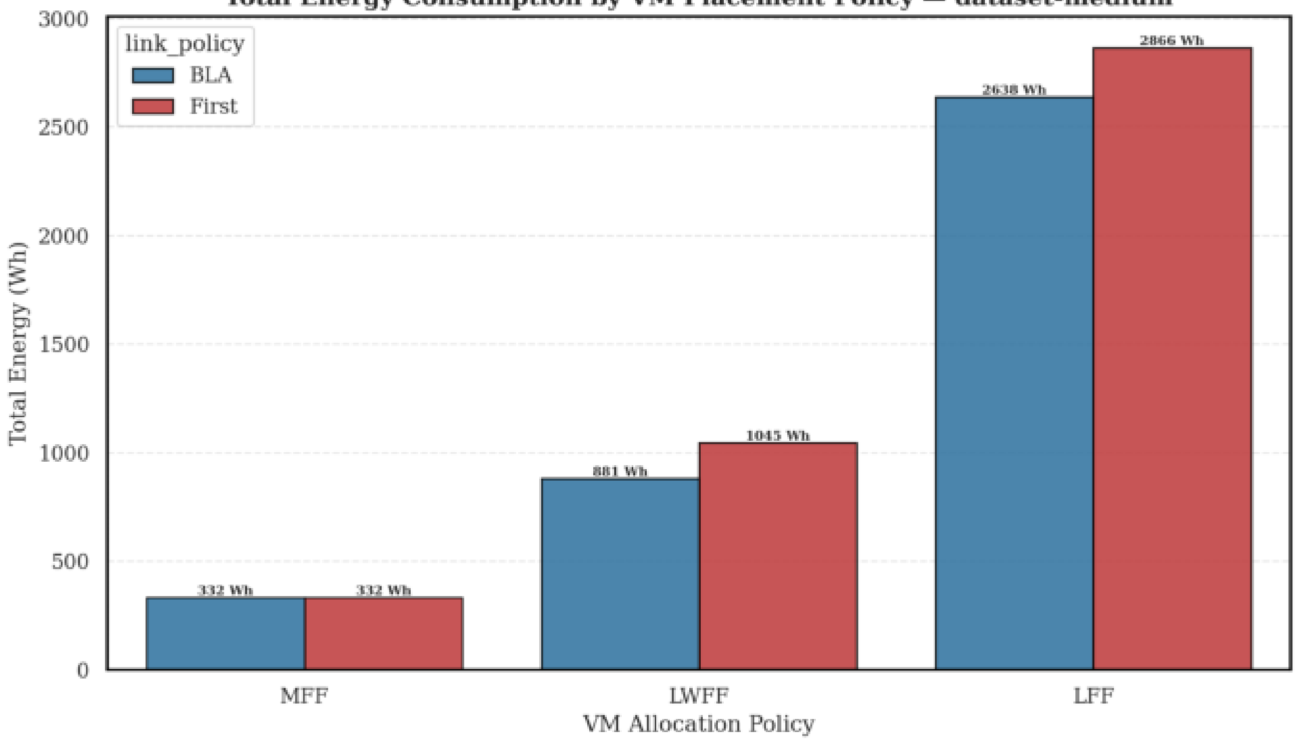
Table 6: Total energy consumption (Wh)

Dataset	VM Policy	SFL (Wh)	BLA (Wh)	Gain
Medium	LWFF	1,045.4	880.8	-15.7%
Medium	LFF	2,865.8	2,638.3	-7.9%

Energy savings grow super-linearly with scale (3.3% -> 6.3% -> 9.2%).

Figure 3: Impact Énergétique (Medium)

irect consequence of reduced simulation duration.



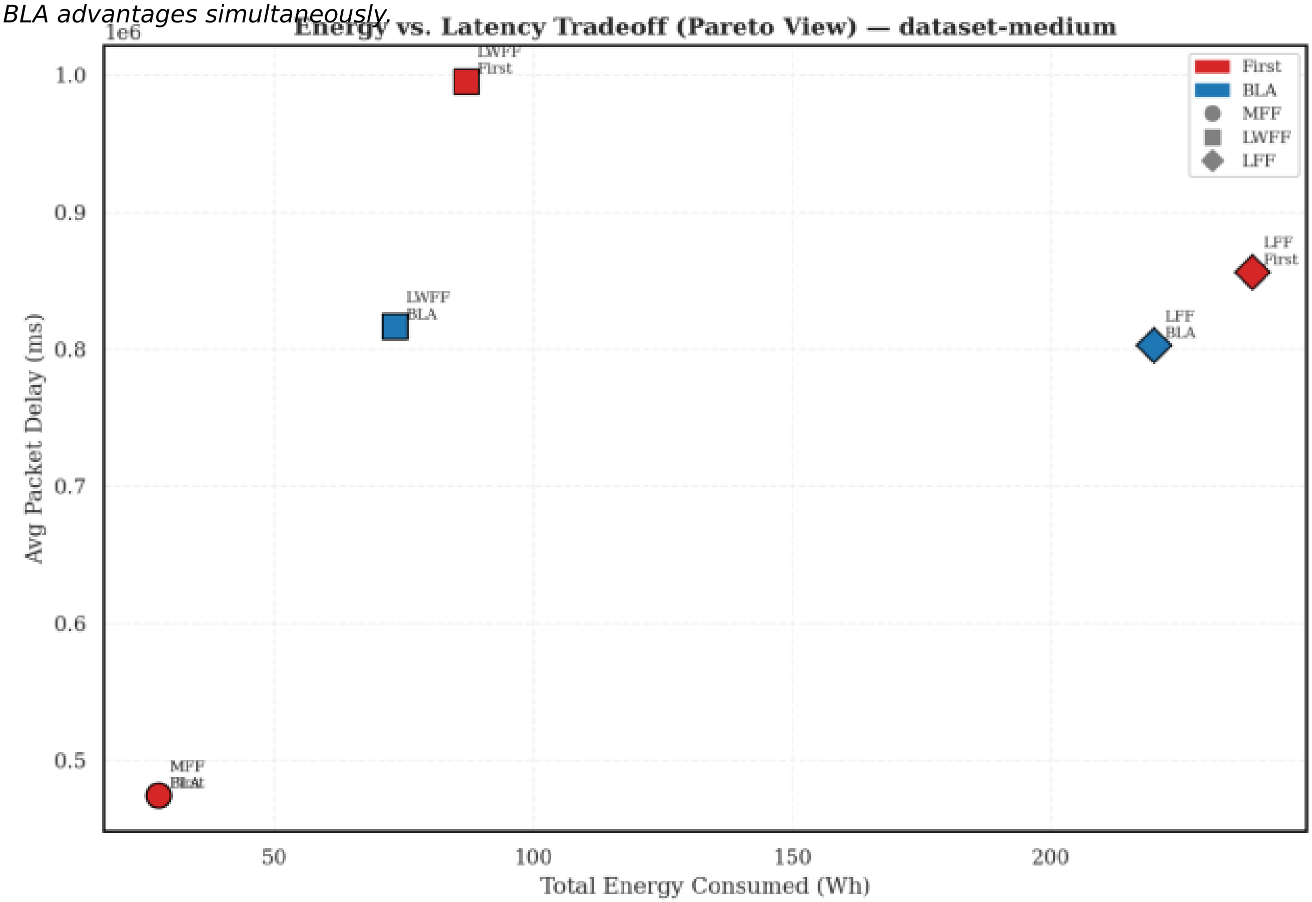
4.2.6 Pareto Analysis

4.2.6 Pareto Analysis: Energy–Latency Trade-off
BLA configurations consistently Pareto-dominate SFL counterparts for LFF and LWFF.
This means BLA achieves lower energy AND lower latency simultaneously.

The clean cluster separation between datasets confirms simulation calibration.
Ordering Mini < Small < Medium is strictly maintained on both axes.

Figure 4: Pareto Trade-off (Medium)

strates both BLA advantages simultaneously.



4.3 Synthesis and Discussion

4.3 Cross-Dataset Synthesis and Scalability

Dataset	Delta Delay	Delta Queue	Delta Energy	Peak SLA gain
Mini	-17.8%	-17.1%	-3.3%	-36.4% (LFF)
Small	-6.0%	-8.9%	-6.3%	-46.4% (LFF)
Medium	-10.0%	-11.7%	-9.2%	-74.9% (LWFF)

4.4 Discussion

BLA provides increasingly stronger QoS protection as congestion intensifies.

MFF produces zero gain across all datasets, acting as a structural internal control.

LWFF+BLA achieves the best balanced outcome across all dimensions.